

PHOTOVOLTAIC BATTERY CHARGE CONTROLLER



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A charge controller is one of the main components of a solar battery charging system (Fig. 1). Its main function is to fully charge a battery without permitting overcharge or reverse current flow (generally during night). If a solar array is connected to a lead-acid battery with no overcharge protection, the battery life is adversely affected.

Ordinary controllers consist of a relay that opens the charging circuit when a preset high-voltage point is reached and closes the circuit again when a preset low-voltage limit is reached, allowing charging to continue. More sophisticated controllers utilise pulse-width modulation (PWM) or maximum power-point tracking (MPPT) to ensure that the battery is being fully charged in the most efficient manner.

The charge controller design described here incorporates the following features:

1. Based on PWM technique
2. Battery temperature compensation
3. Battery charge/discharge mode indication (through LED)
4. Reverse-current protection for the PV module
5. Limited protection against lightning and wrong polarity
6. Over-current protection (fuse and automatic load disconnect)
7. Low-voltage load-disconnect indication
8. Load-on indication (through LED)

PV panel selection

A photovoltaic (PV) panel or module comprises a number of PV cells in various series-parallel combinations in order to generate the required power (voltage×current). Typically,

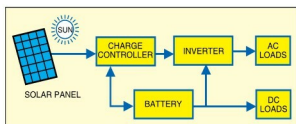


Fig. 1: Block diagram of a typical solar battery charging system

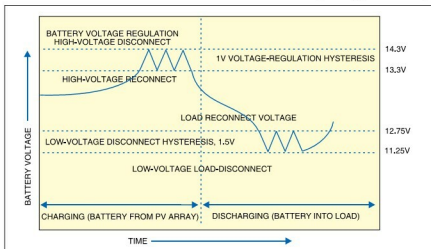


Fig. 2: Charging and discharging set points at 25°C

a PV cell built using a $9 \times 9 \text{ cm}^2$ (81 cm^2) wafer, when subject to sunlight, could produce around 0.5V, which could source around 2.5A of current into a load. Note that a PV cell or module is not damaged when its terminals are shorted. Key specifications of a PV panel are:

P_{MAX} : Maximum power in watts ($P_{\text{MAX}} = V_{\text{MP}} \times I_{\text{MP}}$)

V_{OC} : Open-circuit voltage (maximum voltage with module output open circuit)

V_{MP} : Voltage at which the module produces the highest power (also called operating voltage)

I_{MP} : Current at which the module produces the highest

power (also called operating current)

I_{SC} : Short-circuit current (maximum current with output shorted)

Minimum acceptable V_{MP} voltage rating.

To charge a 12V battery, a PV module with higher V_{MP} in the range

Typical Rating of a PV Module

Electrical parameters	PM80
Maximum power rating P_{MAX} (Wp)*	80.0
Minimum power rating P_{MIN} (Wp)*	75.0
Rated current I_{MP} (A)	4.6
Rated voltage V_{MP} (V)	17.5
Short-circuit current I_{SC} (A)	5.0
Open-circuit voltage V_{OC} (V)	21.5
Physical parameters	Values
Number of cells	36
Physical dimension (mm) (L×W×T)	1200×550×35
Weight (kg)	7.5

Note: 1. There will be 8-10 per cent increase in power after lamination using high-transmission iron-free glass and ethylene vinyl acetate (EVA).

2. Standard test condition (STC)—Solar irradiation: AM 1.5; Intensity: 100 milliwatts/cm²; Cell temperature: 25°C (within the measurement tolerance of ±5 per cent)